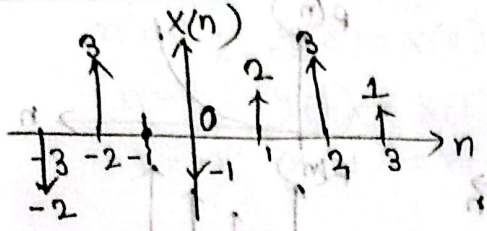
DSP Lec-02

Representation of DTS (Discrete time signal):-

1. Graphical 2. Functional 3. Tabular 4. Sequence

Graphical



$x(-3) = -2$ $x(1) = 2$

$x(-2) = 3$ $x(2) = 3$

$x(-1) = 0$ $x(3) = 1$

$x(0) = -1$

Functional

$$x(n) = \begin{cases} -2 & \text{for } n = -3 \\ 3 & \text{for } n = -2 \\ 0 & \text{for } n = -1 \\ -1 & \text{for } n = 0 \\ 2 & \text{for } n = 1 \\ 3 & \text{for } n = 2 \\ 1 & \text{for } n = 3 \end{cases}$$

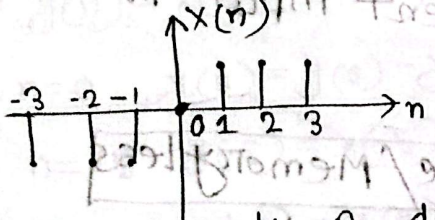
Tabular:

n	-3	-2	-1	0	1	2	3
x(n)	-2	3	0	-1	2	3	1

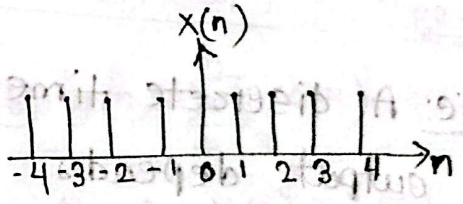
Sequence :- $x(n) = \{-2, 3, 0, -1, 2, 3, 1\}$

Odd signal : A signal is said to be odd if it satisfied the following condition:-

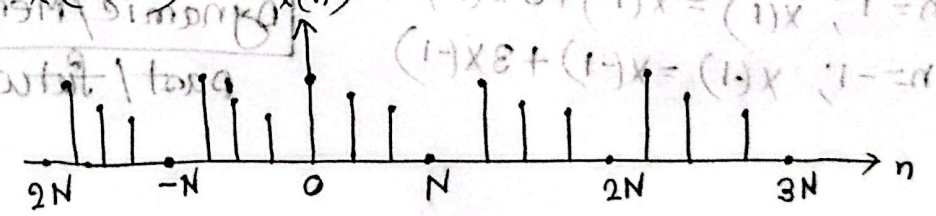
$x(n) = -x(-n)$



Even Signal:- A signal is said to be even or symmetric if it satisfied the following condition $x(n) = x(-n)$



Periodic Signal: A discrete time signal is periodic if $x(n) = x(n+N)$



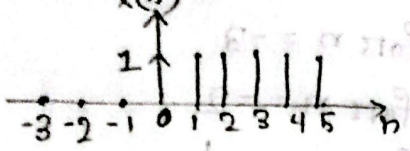
Aperiodic
$x(n) \neq x(n+N)$

Elementary discrete time signals:-

It calls building blocks for more complex signal.

Unit step:

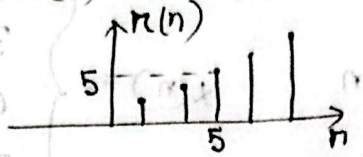
$$u(n) = \begin{cases} 1 ; & n \geq 0 \\ 0 ; & n < 0 \end{cases}$$



Unit Ramp

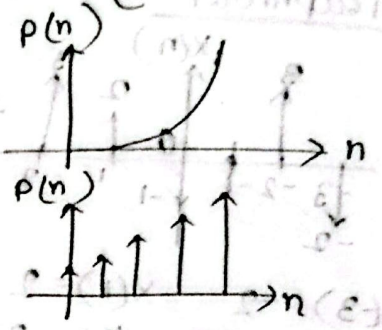
$$r(n) = \begin{cases} n ; & n \geq 0 \\ 0 ; & n < 0 \end{cases}$$

$r(n) = n\,u(n)$



Unit parcabolic

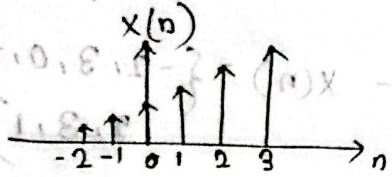
$$p(n) = \begin{cases} n^2/2 \text{ for } n \geq 0 \end{cases}$$



Real Exponential Sequence:-

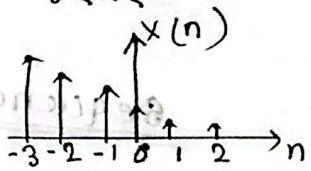
$x(n) = a^n$ for all n

a > 1



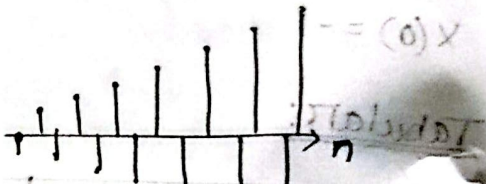
Exponentially grows

0 < a < 1



exponentially decays

a < 0



taking alternate sign

Lec-03 **Classification of discrete time signal**

Static: A discrete time system is called static if its output depends only present input is called static.

$$x(n) = x(n) + 3x(n)$$

$n = 0;\ x(0) = x(0) + 3x(0)$

$n = 1;\ x(1) = x(1) + 3x(1)$

$n = -1;\ x(-1) = x(-1) + 3x(-1)$

Static / Memoryless

present

Dynamic / Memory

past / future

Dynamic: A discrete time system is called dynamic if its output depends on the past, future values is called dynamic system/memory.

$$y(n) = x(n) + x(n+6)$$

$n=0$; $y(0) = x(0) + x(6)$

$n=1$; $y(1) = x(1) + x(7)$

$n=-1$; $y(-1) = x(-1) + x(+5)$

Causal / Non causal

↓ past + present ↓ future

Dynamic / static:

(a) $y(n) = x(n+2)$

$n=0$ $y(0) = x(2)$

$n=5$ $y(5) = x(7)$

$n=-4$ $y(-4) = x(-2)$

dynamic

(b) $y(n) = x^2(n)$

$n=0$ $y(0) = x^2(0)$

$n=5$ $y(5) = x^2(5)$

$n=-4$ $y(-4) = x^2(-4)$

static

(c) $y(n) = x(n-2) + x(n)$

$n=0$;

$y(0) = x(-2) + x(0)$

Dynamic

Causal / Non causal:

(a) $y(n) = x(n) - x(n-1)$

$n=0$ $y(0) = y(0) - y(-1)$

$n=-5$ $y(-5) = x(-5) - x(-6)$

$n=4$ $y(-4) = x(4) - x(3)$

Causal

(b) $y(n) = \sum_{k=-\infty}^{n} x(k)$

$k=0$ $y(0) = x(0)$ ①

$k=-1$ $y(-1) = x(-1)$

$k=+4$ $y(4) = x(4)$

Causal

e $y(n) = ax(n)$

$n=0 \quad y(0) = ax(0)$

$n=-5 \quad y(-5) = ax(-5)$

$n=+4 \quad y(+4) = ax(4)$

Causal

d $y(n) = x(n) + 3x(n+4)$

$n=0 \quad y(0) = x(0) + 3x(4)$

$n=5 \quad y(5) = x(5) + 3x(9)$

$n=-3 \quad y(-3) = x(-3) + 3x(1)$

Non causal

c $y(n) = x(n^2)$

$n=0 \quad y(0) = x(0)$

$n=5 \quad y(5) = x(25)$

$n=-4 \quad y(-4) = x(16)$

Non causal

f $y(n) = x(2n)$

$n=0 \quad y(0) = x(0)$

$n=5 \quad y(5) = x(10)$

$n=-3 \quad y(-3) = x(-6)$

Non causal

g $y(n) = x(-n)$

$n=0 \quad y(0) = x(-0)$

$n=5 \quad y(5) = x(-5)$

$n=-3 \quad y(-3) = x(+3)$

Non causal

Time Invariant

A system is called ~~to be~~ time invariant if its input output characteristics doesn't change with time.

$x(n) \longrightarrow y(n)$

$x(n-n_0) \longrightarrow y(n-n_0)$

TIV $\Rightarrow x(n-n_0) = y(n-n_0)$

~~TV~~ TV $\Rightarrow x(n-n_0) \neq y(n-n_0)$

input → delay $x(n)$

output → ~~sig~~ n delay

① $y(n) = x(n) + x(n-1)$

$y'(n) = x(n-n_0)$

$= x(n-n_0) + x(n-n_0-1)$

$y''(n) = y(n-n_0) = x(n-n_0) + x(n-n_0-1)$

$y'(n) = y''(n)$ TIV

(ii) $y(n) = n\,x(n)$

$y'(n) = x(n-n_0) = n\,x(n-n_0)$

$y''(n) = y(n-n_0) = (n-n_0)\,x(n-n_0)$

$y'(n) \neq y''(n)$ TV

(iii) $y(n) = x(2n)$

$y'(n) = x(n-n_0) = x(2n-n_0)$

$y''(n) = y(n-n_0) = x(2n-2n_0)$

$y'(n) \neq y''(n)$ TV

(iv) $y(n) = 3x(n)$

$y'(n) = x(n-n_0)$

$= 3x(n-n_0)$

$y''(n) = y(n-n_0)$

$= 3x(n-n_0)$

$y'(n) = y''(n)$ TI

A linear system is one that satisfies the Superposition theorem. It has two property:-

Additivity: $x_1(n) \longrightarrow y_1(n)$

$x_2(n) \longrightarrow y_2(n)$

$x_1(n) + x_2(n) \longrightarrow y_1(n) + y_2(n)$

Homogenity: $x(n) \longrightarrow y(n)$

$a\,x(n) \longrightarrow a\,y(n)$

A system is linear if

$$T\{a_1x_1(n) + a_2x_2(n)\} = a_1y_1(n) + a_2y_2(n)$$

Linear/Non Linear

(i) $y(n) = n\,x(n)$

$y_1(n) = n\,x_1(n)$

$y_2(n) = n\,x_2(n)$

A linear system holds: -

$T\{a_1x_1(n) + a_2x_2(n)\} \neq a_1y_1(n) + a_2y_2(n)$

$$L.H.S = T\{a_1 x_1(n) + a_2 x_2(n)\}$$

$$= n\{a_1 x_1(n) + a_2 x_2(n)\}$$

$$= n a_1 x_1(n) + n a_2 x_2(n)$$

$$R.H.S = a_1 y_1 + a_2 y_2$$

$$= a_1 n x_1(n) + a_2 n x_2(n)$$

$L.H.S = R.H.S \longrightarrow$ Linear

(ii) $y(n) = x(n^2)$

$y_1(n) = x_1(n^2)$

$y_2(n) = x_2(n^2)$

$$T\{a_1 x_1(n) + a_2 x_2(n)\} = a_1 y_1(n) + a_2 y_2(n)$$

$$L.H.S = T\{a_1 x_1(n) + a_2 x_2(n)\}$$

$$= \{a_1 x_1(n^2) + a_2 x_2(n^2)\}$$

$$= a_1 x_1(n^2) + a_2 x_2(n^2)$$

$$R.H.S = a_1 y_1(n) + a_2 y_2(n)$$

$$= a_1 x_1(n^2) + a_2 x_2(n^2)$$

$L.H.S = R.H.S \longrightarrow$ Linear

(iii) $y(n) = x^2(n)$

$y_1(n) = x_1^2(n)$

$y_2(n) = x_2^2(n)$

$T\{a_1x_1(n) + a_2x_2(n)\} = a_1y_1(n) + a_2y_2(n)$

$L.H.S = T\{a_1x_1(n) + a_2x_2(n)\}$

$= \{a_1x_1(n) + a_2x_2(n)\}^2$

$R.H.S = a_1y_1(n) + a_2y_2(n)$

$= a_1x_1^2(n) + a_2x_2^2(n)$

$L.H.S = R.H.S \longrightarrow$ Non Linear

(iv) $y(n) = Ax(n) + B$

$y_1(n) = Ax_1(n) + B$

$y_2(n) = Ax_2(n) + B$

$T\{a_1x_1(n) + a_2x_2(n)\} = a_1y_1(n) + a_2y_2(n)$

$L.H.S = T\{a_1x_1(n) + a_2x_2(n)\}$

$= Aa_1x_1(n) + Aa_2x_2(n) + B$

$R.H.S = a_1Ax_1(n) + B(a_1 + a_2) + a_2Ax_2(n)$

$L.H.S \neq R.H.S \longrightarrow$ Non Linear

⑥ $y(n) = e^{x(n)}$

$y_1(n) = e^{x_1(n)}$

$y_2(n) = e^{x_2(n)}$

$T\{a_1x_1(n) + a_2x_2(n)\} = a_1y_1(n) + a_2y_2(n)$

$L.H.S = T\{a_1x_1(n) + a_2x_2(n)\}$

$= e^{a_1x_1(n) + a_2x_2(n)}$

$= e^{a_1x_1(n)} \cdot e^{a_2x_2(n)}$

$R.H.S = a_1y_1(n) + a_2y_2(n)$

$= a_1 e^{x_1(n)} + a_2 e^{x_2(n)}$

$L.H.S \neq R.H.S \Longrightarrow$ Non Linear

Stable System: A stable system satisfied the Bounded Input Bounded output (BIBO) condition

Here bounded means infinite amplitude.

For a stable system output should be bounded or finite.

for finite/bounded input at every instant of time

- $|x(n)| \leq M_x \leq \infty$
- $|x(n)| \leq M_y < \infty$

Ex: $\sin(n), \cos(n), u(n)$

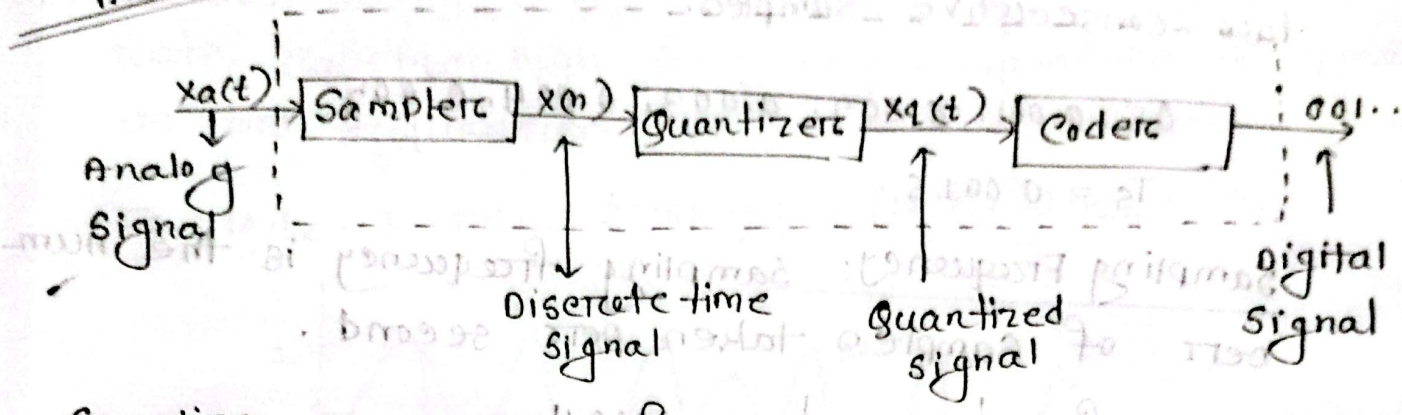
M_x = Maximum absolute value of Input signal

M_y = " " " " Output signal

Analog to Digital Conversion

lec-04

Steps of ADC



Sampling: Conversion of continuous time signal into discrete time signal by taking samples at discrete time instant.

Quantization: Conversion of Discrete time continuous valued signal into discrete time discrete valued signal.

Coding: Each discrete value is represented by binary bit sequence.

Sampling

Sampling is the process of continuous time signal in discrete time signal.

Mathmatically $x[n] = x(nT_s)$ T_s = Sampling Interval

$$T_s = \frac{1}{f_s}$$

Sampling frequency $f_s = \frac{1}{T_s}$

Sampling Interval:- It is the gap between the two consecutive samples.

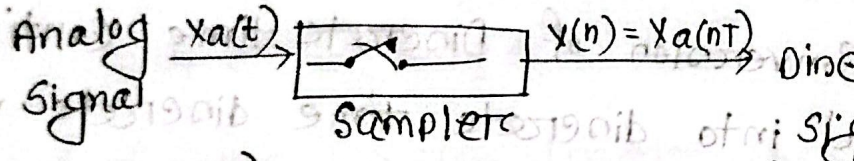
0s, 0.001, 0.002, 0.003, 0.004, 0.005 ...

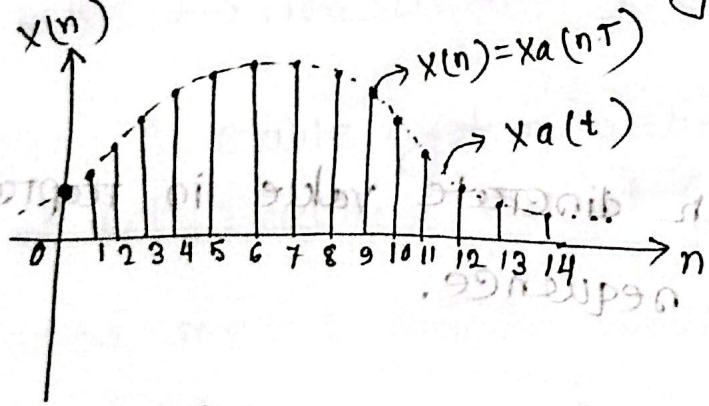
$T_s = 0.001s$

Sampling Frequency: Sampling frequency is the number of samples taken per second.

$$f_s = \frac{1}{T_s} = \frac{1}{0.001} = 1000\ Hz$$

We take 1000 samples per second.





$x(t) = A\cos(2\pi f t)$ — f = original/given frequency in equation

$= A\cos(2\pi f n T_s)$

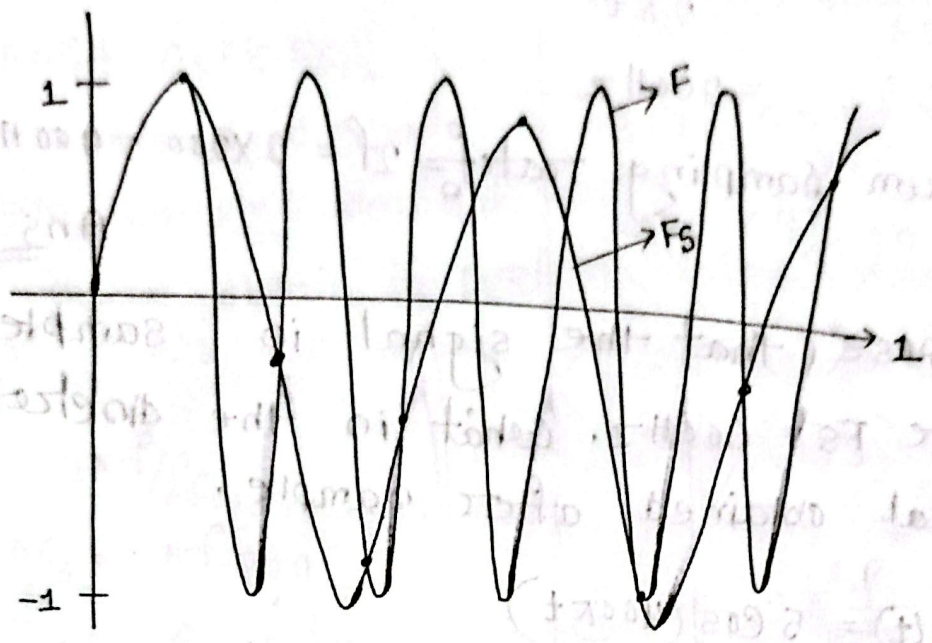
$= A\cos\left(2\pi f n \frac{1}{f_s}\right)$ — f_s = Sampling frequency in question no.

$= A\cos\left(2\pi \frac{f}{f_s} n\right)$

$= A\cos(2\pi F n)$

Aliasing Signal (Fake Signal): The distortion that occurs when a signal sampled below Nyquist rate, causing high frequency components to appear as low frequencies.

Fs = 10Hz, F = 8Hz, Sampling = (10-8) = 2Hz



3 Characteristics of Sampling:-

① Nyquist Theorem: If a signal can be constructed perfectly then it has ~~two~~ to be sampled at least twice its highest frequency $f_s \geq 2 f_{max}$

② Nyquist rate: The minimum sampling rate required, equal to the 2x highest frequency of signal

$f_s \geq 2f$ Let $f_s = 16Hz$ $f = 8Hz$

$16 \geq 16$

③ Aliasing | 8Hz sampling signal is looking like original signal

Slide 16

$x_a(t) = 5\cos(400\pi t)$

(a) Determine the minimum sampling rate required to avoid aliasing.

$2\pi f t = 400\pi t$

$f = \frac{400\pi t}{2\pi t}$

$= 200\text{Hz}$

Minimum sampling rate $f_0 = 2f = 2 \times 200 = 400\text{Hz}$

Ans

b) Suppose that the signal is sampled at rate $F_s = 600\text{Hz}$. What is the discrete time signal obtained after sample.

$x_a(t) = 5\cos(400\pi t)$

$= 5\cos(2\pi\, 200\, t)$

$= 5\cos\left(2\pi\, 200 \times \frac{nT_s}{}\right)$

$= 5\cos\left(2\pi\, 200\, \frac{1}{F_s}\right)n$

$= 5\cos\left(2\pi \frac{200}{600_3}\right)n$

$= 5\cos(2\pi/3)n$

c) $F_s = 300\text{Hz}$

$x_a(t) = 5\cos(400\pi t)$

$= 5\cos(2\pi\, 200\, nT_s)$

$= 5\cos(2\pi\, 200/F_s)n$

$= 5\cos(2\pi\, 200/300)n$

$= 5\cos(4\pi/3)n$

$= 5\cos(4\pi/3 - 2\pi)n$

$= 5\cos\left(\frac{4\pi - 6\pi}{3}\right) n$

$= 5\cos(-2\pi/3)\, n$

$= 5\cos(2\pi/3)\, n$

Ans

(d) What is the frequency $0 < F < F_s/2$ of a sinusoid that yields samples identical to those obtain in part (c)?

$\omega = 2\pi f/f_s$

$\Rightarrow 2\pi/3 = 2\pi f/300$

$\Rightarrow 300/3 = f = 100\,Hz$

Ans

slide 15

$x_a(t) = 3\cos 100\pi t$

(a) $2\pi f t = 100\pi t$

$f = 50\,Hz$

Min sampling rate $f_s = 2f$

$= 100\,Hz$

Ans

(b) $x_a(t) = 3\cos(100\pi t)$

$= 3\cos(2\pi 50 n T_s)$

$= 3\cos\left(2\pi \frac{50}{F_s}\right) n$

$= 3\cos\left(2\pi \frac{50}{200}\right) n$

$= 3\cos\left(\frac{\pi}{2}\right) n$

Ans

(c) $x_a(t) = 3\cos(100\pi t)$

$= 3\cos(2\pi 50 n T_s)$

$= 3\cos\left(2\pi 50 \frac{1}{F_s}\right) n$

$= 3\cos\left(\frac{2\pi 50}{75}\right) n$

$= 3\cos\left(\frac{4\pi}{3}\right) n$

$= 3\cos(4\pi/3 - 2\pi)\, n$

$= 3\cos(2\pi/3)\, n$

Ans

(d) $\omega = 2\pi \frac{f}{f_s}$

$\Rightarrow 2\pi/3 = 2\pi \frac{f}{75}$

$\Rightarrow f = 25\,Hz$

Ans

Quantization

Quantization means rounding the continuous amplitude of each samples set to the nearest fixed level.

Why do we need it? → Because digital system can only store and process a finite no of different values.

Types of Quantizers:

1. Uniform Quantizers:-

(i) All quantization levels are equally spaced

(ii) Step size (Δ) is constant

(iii) Most commonly used in DSP (PCM systems)

2. Non uniform Quantizers:-

(i) Step size is not constant

(ii) Smaller steps near zero, larger steps for high amplitude.

(iii) Used for speech/audio.

Two Common Uniform Quantizer Types:-

Mid tread → zero is exactly at a level

Mid Riser → zero is between two levels.

Dynamic Range:- It is the signal that is the difference between its maximum and minimum amplitude values.

Dynamic Range (DR) = $X_{max} - X_{min}$

Quantization Levels (L):- The no of discrete levels the analog amplitude is divided into is denoted as L. For a 3-bit ADC $L = 2^b = 2^3 = 8$

So, the analog signal will be divided into 8 discrete levels.

Step Size/Resolution (Δ):- The step size is the smallest change in amplitude that can be defined as:

$$\Delta = \frac{X_{max} - X_{min}}{L} = \frac{\text{Dynamic Range}}{\text{No of Level.}}$$

Quantization Error: Quantization error can be defined as the difference between original samples and quantized sample.

$$e[n] = X[n] - Xq[n] \qquad e[n] \rightarrow \pm \Delta/2$$

$X[n]$ ↓ original sample, $Xq[n]$ → Quantized sample

Coding

After sampling and quantization, we get discrete amplitude levels.

SNR: a general measure of how strong my desired signal is compared to the unwanted noise corrupting it.

General SNR $= 10\log_{10}\left(\frac{P_{signal}}{P_{Noise}}\right)$ dB

SQNR (SNR Quantization specific)

$= 10\log_{10}\left(\frac{P_{signal}}{P_{Noise}}\right)$ dB $= (6.02 \times b + 1.76)$ dB

derived assuming sinusoidal signal, uniform quantizer.

In DSP SNR and SQNR are same because noise source is only quantization.

Higher SQNR → better quality

PB1 A uniform quantizer has a signal range of −4V to +4V and uses 3 bit

(a) $L = 2^3 = 8$ (b) $\Delta = \frac{4+(4)}{L} = 1$ V

(c) Max error $= \pm \Delta/2 = \pm 0.5$ V Ans

PB2 Calculate the SQNR for a 4-bit uniform quantizer SQNR $= (6.02 \times 4 + 1.76)$ dB $= 25.84$ dB

Ans

PB3 An audio application required SQNR of at least 72 dB. How many bits are needed, what is L?

SQNR $= 6.02 \times b + 1.76$

$72 = 6.02 \times b + 1.76$

$b = 11.66 \approx 12$ bits

$L = 2^b = 4096$ levels

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General SNR $= 10\log_{10}\left(\frac{P_{signal}}{P_{Noise}}\right)$ dB

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(a) $L = 2^3 = 8$ (b) $\Delta = \frac{4+(4)}{L} = 1V$

(c) Max error $= \pm \Delta/2 = \pm 0.5V$ Ans

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$b = 11.66 \approx 12$ bits

$L = 2^b = 4096$ levels

Pb4 Compare 3 bit and 8 bit quantizers for each, compute SQNR, no of levels, step size (±1V). By how many dB does the 8 bit system improve over 3 bit.

$b = 3$

$L = 2^3 = 8$

$\Delta = \frac{1+1\ (2)}{L\ (8)}$

$= 0.25\ V$

$SQNR = (6.02 \times 3 + 1.76)\ dB$

$= 19.82\ dB$

$b = 8$

$L = 2^8 = 256$

$\Delta = \frac{2}{256} = 0.0078\ V$

$SQNR = (6.02 \times 8 + 1.76)\ dB$

$= 49.92\ dB$

Improvement $= (49.92 - 19.82) = 30.1\ dB$

Extra bits $= \frac{30.1}{6.02} = 5$ Ans

Pb5 An ECG sensor measured signals in the range ±5mV. A doctor required quantization error to never exceed 0.02 mV. Determine the minimum b and L.

$e = \pm \Delta/2 \leq 0.02mV$

$\Rightarrow \Delta \leq 0.04\ mV$

$\Delta = \frac{10}{2^b}$

$L \geq range/\Delta = 10mV/0.04mV$

$= 250$

$L = 256$ $b = 8$ bits

Lec-05 LTI System

A LTI system satisfies the following two criteria:

Linearity: The system obeys superposition theorem

Time Invariance: Characteristics doesn't change with time.

Impulse system is a system that defines how does a system react to a single instant shock

$$h(n) = \delta(n) = \begin{cases} 1 ; n = 0 \\ 0 ; n \neq 0 \end{cases}$$

For any LTI system $y(n) = x(n) * h(n)$

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)\, h(n-k)$$

→ Convolution

Convolution: It is a mathmatical operation that combine the input signal with the impulse response to produces the output of an LTI system.

Steps in Convolution:

Folding: Fold $h(k)$ from $k=0$ to obtain $h(-k)$

Shifting: shift to the $h(-k)$ by n to the right obtain $h(n-k)$

Multiplication: Multiply by $x(k)$ with $h(n-k)$ to obtain $x(k)\, h(n-k)$

Summation: Sum all the values of n

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)\, h(n-k)$$

lec-05 LTI System

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$$y(n) = \sum_{k=-\infty}^{\infty} x(k)\, h(n-k)$$

Math: 1. $x(n) = [1, 2]$ $h(n) = [1, 2]$

Start $= 0+0 = 0$ End $= 1+1 = 2$

$L = (2+2)-1 = 4-1 = 3$ $y(0), y(1), y(2)$

y(0):

$k=0$; $x(0)\,h(0-0) = x(0)\,h(0) = 1 \times 1 = 1$

$k=1$; $x(1)\,h(0-1) = x(1)\,h(-1) = 0 \rightarrow$ SUM $= 1+0 = 1$

y(1):

$k=0$. $x(0)\,h(1-0) = x(0)\,h(1) = 1 \times 2 = 2$

$k=1$; $x(1)\,h(0) = 2 \times 1 = 2 \rightarrow$ SUM $= 4$

y(2):

$k=0$; $x(0)\,h(2) = 0$

$k=1$; $x(1)\,h(1) = 4 \rightarrow$ SUM $= 4$

$y(n) = \{1, 4, 4\}$ (↑ 1)

Ans

[2] $x(n) = [1, 2, 3]$ (↑ 2) $h(n) = [1, 2, 1]$

Start $= -1+0 = -1$ end $= 1+2 = 3$

$L = (3+3)-1 = 5$ $y(-1), y(0), y(1), y(2), y(3)$

y(-1):

$k=-1$; $x(-1)\,h(-1+1) = x(-1)\,h(0) = 1 \times 1 = 1$

$k=0$; $x(0)\,h(-1-0) = x(0)\,h(-1) = 0$ $\rightarrow$ SUM $= 1$

$k=1$; $x(1)\,h(-1-1) = x(1)\,h(-2) = 0$

$y(0)$:

$k=-1$; $x(-1)\ h(0-(-1)) = 1 \times 2 = 2$

$k=0$; $x(0)\ h(0-0) = 2 \times 1 = 2 \rightarrow SUM = 4$

$k=1$; $x(1)\ h(0-1) = 3 \times 0 = 0$

$y(1)$:

$k=-1$; $x(-1)\ h(1+1) = 1 \times 1 = 1$

$k=0$; $x(0)\ h(1-0) = 2 \times 2 = 4 \rightarrow SUM = 8$

$k=1$; $x(1)\ h(0) = 3 \times 1 = 3$

$y(2)$:

$k=-1$; $x(-1)\ h(3) = 1 \times 0 = 0$

$k=0$; $x(0)\ h(2) = 2 \times 1 = 2 \rightarrow SUM = 8$

$k=1$; $x(1)\ h(1) = 3 \times 2 = 6$

$y(3)$:

$k=-1$; $x(-1)\ h(4) = 1 \times 0 = 0$

$k=0$; $x(0)\ h(3) = 1 \times 0 = 0$

$k=1$; $x(1)\ h(2) = 3 \times 1 = 3 \rightarrow SUM = 3$

$y(n) = \{1, 4, 8, 8, 3\}$ (↑ at 4)

Ans

Correlation

- Cross Correlation
  - Measures the similarities between to different signal
- Auto Correlation
  - Measures the Similarity between with itself

Cross $\rightarrow r_{xy}(n) = \sum_{k=-\infty}^{\infty} x(k)\, y(k+n)$

Auto: $r_{xx}(n) = \sum_{k=-\infty}^{\infty} x(k)\, x(k+n)$

1. $x(n) = [1, 2]$ $y(n) = [1, 1]$

$r_{xy}(n) = ?$ $L = (4-1) = 3$ start $= 0+0 = 0$, end $= 1+1 = 2$

$r_{xy}(0)$, $r_{xy}(1)$, $r_{xy}(2)$

$r_{xy}(0)$

$k=0$; $x(0)\, y(0) = 1 \times 1 = 1$

$k=1$; $x(1)\, y(1) = 2 \times 1 = 2$ $\rightarrow$ SUM $= 3$

$r_{xy}(1)$

$k=0$; $x(0)\, y(1) = 1 \times 1 = 1$

$k=1$; $x(1)\, y(2) = 2 \times 0 = 0$ $\rightarrow$ SUM $= 1$

$r_{xy}(2)$

$k=0$; $x(0)\, y(2) = 1 \times 0 = 0$

$k=1$; $x(1)\, y(3) = 2 \times 0 = 0$ $\rightarrow$ SUM $= 0$

$r_{xy} = \{3, 1, 0\}$ (3 at the origin)

2. $x(n) = \{3, 2\}$ (3 at the origin) $x(n) = [3, 2]$ (2 at the origin)

start $= -1-1 = -2$, end $= 0+0 = 0$, $L = 4-1 = 3$

$r_{xx}(-2)$, $r_{xx}(-1)$, $r_{xx}(0)$

$r_{xx}(-2)$

$k=-1$; $x(-1)\, x(-3) = 3 \times 0 = 0$

$k=0$; $x(0)\, x(-2) = 2 \times 0 = 0$

2. $h(n) = [3, \underset{\uparrow}{2}]$ $x(n) = [\underset{\uparrow}{2}, 3]$

Start = $-1+0 = -1$ End = $0+1 = 1$ $L = (4-1) = 3$

$r_{xx}(-1)$, $r_{xx}(0)$, $r_{xx}(1)$

$r_{xx}(-1)$:

$k = 1$; $x(-1)\,x(-2) = 3 \times 0 = 0$

$k = 0$; $x(0)\,x(-1) = 3 \times 2 = 6 \rightarrow$ SUM = 6

$r_{xx}(+0)$:

$k = -1$; $x(-1)\,x(-1) = 3 \times 3 = 9$

$k = 0$; $x(0)\,x(0) = 2 \times 2 = 4 \rightarrow$ SUM = 13

$r_{xx}(1)$

$k = -1$ $x(-1)\,x(0) = 3 \times 2 = 6$

$k = 0$; $x(0)\,x(1) = 2 \times 0 = 0 \rightarrow$ SUM = 6

$r_{xx}(n) = \{6, \underset{\uparrow}{13}, 6\}$

lec-06 z-Transform

z-transform: It converts a discrete time signal from time domain into complex frequency domain

Why to use z-transform:-

(i) Converts discrete time difference equation into simple algebraic equations.

(ii) Turn convolution in time domain into multiplication in z-domain.

(iii) Helps analyze system properties like, stability and causality using poles and ROC.

Application of z transform

i) Digital Audio Processing :-
- Design of digital filters, audio effects, noise reduction.

ii) Image Processing :-
- Edge detection, image enhancement, compression.

iii) Control System :-
- Stability analysis and controllers design for discrete time systems.

iv) Biomedical signal Processing :-
- Analysis and filtering of ECGs, EEGs, and other biological signals for diagnosis.

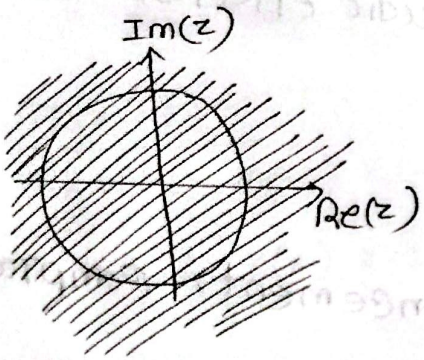
v) Communication system :-
- Modulation, demodulation, error correction, etc.

Region of Convergence (ROC): The ROC is the set of all values of z for which the z transform converges.

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

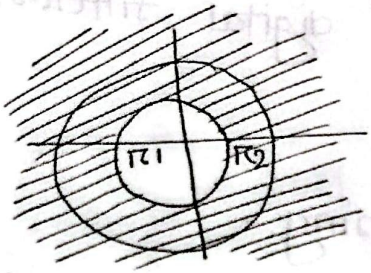
Case: I

ROC: all z



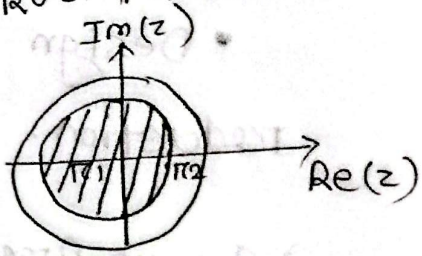
Case: II

ROC: $|z| > r_1$

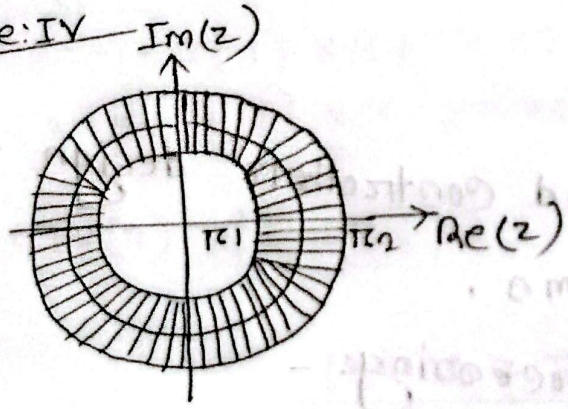


Case: III

ROC: $|z| < r_2$



Case: IV



Ex: 3.1.1

$$x(n) = \{1, 2, 5, 7, 0, 1\}$$

(1 at $n=0$, 1 at $n=5$)

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} = \sum_{n=0}^{\infty 5} x(n) z^{-n}$$

$$= x(0)z^{-0} + x(1)z^{-1} + x(2)z^{-2} + x(3)z^{-3} + x(4)z^{-4} + x(5)z^{-5}$$

$$= 1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + z^{-5}$$

ROC: ROC is everywhere except $z = 0$

Ex: 02

$x(n) = \{1, 2, \underset{\uparrow}{3}, 4, 0, 5\}$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$= \sum_{n=-2}^{3} x(n) z^{-n}$$

$$= x(-2) z^{2} + x(-1) z^{1} + x(0) z^{0} + x(1) z^{-1} + x(2) z^{-2} + x(3) z^{-3}$$

$$= z^{2} + 2z + 3 + 4z^{-1} + 5z^{-3}$$

RoC: RoC is everywhere except $z = 0$ and $z = \infty$

Ex: 3 $x(n) = a^n u(n)$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$= \sum_{n=0}^{\infty} a^n z^{-n}$$

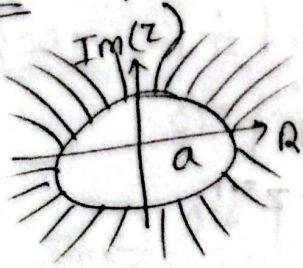
$$= a^0 z^{-0} + a^1 z^{-1} + a^2 z^{-2} + a^3 z^{-3} + \cdots$$

$$= 1 + az^{-1} + a^2 z^{-2} + a^3 z^{-3} + \cdots$$

$$= \frac{1}{1 - az^{-1}}$$

RoC: $|az^{-1}| < 1$ or $|a/z| < 1$ or $|a| < |z|$

or $|z| > a$



ex:4. $x(n) = \delta(n)$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} \delta(n) z^{-n}$$

$\delta(n) = 1$; only when $n = 0$

$$X(z) = \delta(0) z^{-0} = 1$$

ROC: Entire z-plane

Ex:5

$x(n) = \delta(n-k)$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} \delta(n-k) z^{-n}$$

$\delta(n-k) = 1$ only when

$\delta(0) = 1$

$n - k = 0$

$\Rightarrow n = k$

$$X(z) = \delta(k-k) z^{-k}$$

$$= z^{-k}$$

ROC: Entire z-plane except $z = 0$

Ex:6 $x(n) = \delta(n+k)$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} \delta(n+k) z^{-n}$$

$\delta(n+k) = 1$ only when $n + k = 0$

$\Rightarrow n = -k$

$$X(z) = \delta(-k+k) z^{+k} = z^{k}$$

ROC: Entire z-plane except $z = \infty$

Ans

Ex: 3.1.4

$$x(n) = -a^n u(-n-1) = \begin{cases} 0 & ; n \geq 0 \\ -a^n & ; n \leq -1 \end{cases}$$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$= \sum_{n=-\infty}^{-1} x(n) z^{-n} + \sum_{n=0}^{\infty} x(n) z^{-n}$$

$$= \sum_{n=-\infty}^{-1} -a^n u(-n-1) z^{-n} + \sum_{n=-1}^{\infty} -a^n u(-n-1) z^{-n}$$

$$= \sum_{n=-\infty}^{-1} -a^n z^{-n} + 0$$

$$= -\sum_{n=-\infty}^{-1} (az^{-1})^n$$

let $az^{-1} = \pi$

$n = -K$

$K = -\infty \quad n = +\infty$

$K = -1 \quad n = +1$

$$X(z) = -\sum_{n=1}^{\infty} \pi^{-K}$$

$$= -\{\pi^{-1} + \pi^{-2} + \pi^{-3} + \pi^{-4} + \cdots\}$$

$$= -\left\{\frac{1}{\pi}\left\{\frac{1}{\pi} + \frac{1}{\pi^2} + \frac{1}{\pi^3} + \cdots\right\}\right\}$$

$$= -\left\{\frac{1}{\pi} \times \frac{1}{1 - \frac{1}{\pi}}\right\} \quad \text{ROC: } \left|\frac{1}{\pi}\right| < 1$$

$$= -\frac{1}{\pi} \times \frac{\pi}{\pi - 1}$$

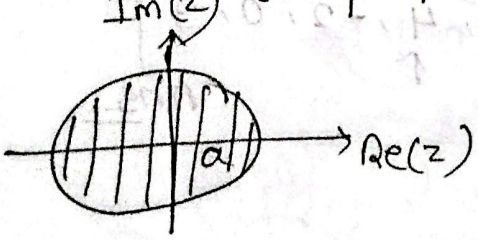
$$= \frac{1}{1 - \pi} = \frac{1}{1 - az^{-1}} = \frac{1}{1 - a/z} = \frac{z}{z - a}$$

Ans

ROC: $\left|\frac{1}{\pi}\right| < 1$

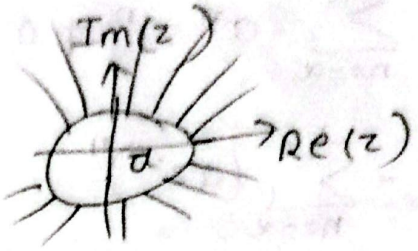
or $\left|\frac{1}{az^{-1}}\right| < 1$

or $|z| < |a|$



3.1.5 $x(n) = a^n u(n) + b^n u(-n-1)$

$x(n) = x_1(n) + x_2(n)$



$X_1(z) = \frac{1}{1-az^{-1}}$; ROC 1 $|z| > |a|$

$X_2(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$

$= \sum_{n=-\infty}^{-1} b^n z^{-n} = \sum_{n=-\infty}^{-1} (bz^{-1})^n$

$u(n-1) = 1$

$-n-1 \geq 0$

$+n \leq -1$

let $bz^{-1} = \pi$ $\quad n = -k$ $\quad n = -\infty$ $\quad k = \infty$

$n = -1$ $\quad k = 1$

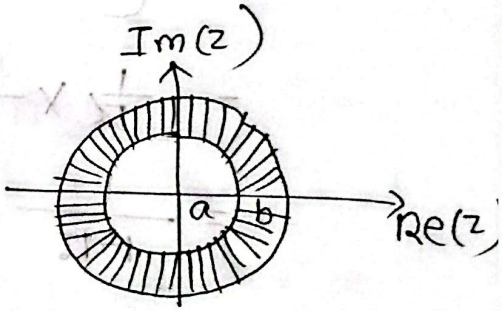
$X_2(z) = \sum_{k=1}^{\infty} \pi^{-k} = \pi^{-1} + \pi^{-2} + \pi^{-3} + \ldots$

$= \frac{1}{\pi}\left\{\frac{1}{\pi} + \frac{1}{\pi^2} + \cdots\right\} = \frac{1}{\pi} \times \frac{1}{1-\frac{1}{\pi}} = \frac{1}{\pi} \times \frac{\pi}{\pi-1}$

$= \frac{1}{\pi-1} = \frac{1}{b/z-1} = \frac{z}{b-z}$

ROC: $|\pi| < 1 \Rightarrow |z/b| < 1 \Rightarrow |z| < |b|$

final ROC: $|a| \leq |z| < |b|$



Inverse z Transform

① $X(z) = 3z^2 + 4 - 2z^{-1} + 3z^{-3}$

$= 3\,\delta(n+2) + 4\,\delta(n) - 2\,\delta(n-1) + 3\,\delta(n-3)$

~~$X(n) = \{3, 4, -2, 3\}$~~

$n = -2 \rightarrow 3$

$n = 0 \rightarrow 4$

$n = 1 \rightarrow -2$

$n = 3 \rightarrow 3$

$X(n) = \{3, 0, \underset{\uparrow}{4}, -2, 0, 3\}$

Ans

Practice

$X(z) = z^2 + 2z^{-1} + 3 + 4z^{-1} + 5z^{-3}$

$x(n) = \delta(n+2) + 2\delta(n-1) + 3\delta(n) + 4\delta(n-1) + 5\delta(n-3)$

$n = -2 \to 1$ $\quad$ $n = 1 \to 4$

$n = 1 \to 2$ $\quad$ $n = 3 \to 5$

$n = 0 \to 3$

$x(n) = \{1, 0, \underset{\uparrow}{3}, 4, 0, 5\}$

Ans

Properties of Linearity:

$x_1(n) \xleftrightarrow{z} X_1(z)$

$x_2(n) \xleftrightarrow{} X_2(z)$

$x(n) = a_1 x_1(n) + a_2 x_2(n)$

$X(z) = a_1 X_1(z) + a_2 X_2(z)$

3.2.1

$x(n) = [3(2^n) - 4(3^n)]u(n)$

$= 3(2^n)u(n) - 4(3^n)u(n)$

~~$= X_1(z) - X_2$~~

Let $a = 2^n$, $a^n = 2^n$

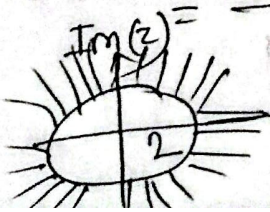
$X_1(z) = 3\sum_{n=-\alpha}^{\alpha} a^n u(n) z^{-n}$

$= 3\sum_{n=0}^{\alpha} a^n z^{-n} = a^0 z^0 + a^1 z^{-1} + a^2 z^{-2} + \cdots$

$= 3(1 + a^1 z^{-1} + a^2 z^{-1} + \cdots)$

$X(z) = \frac{3}{1 - az^{-1}} = \frac{3}{1 - 2z^{-1}}$ $\quad$ ROC: $|z| > |a|$

$|z| > |2|$



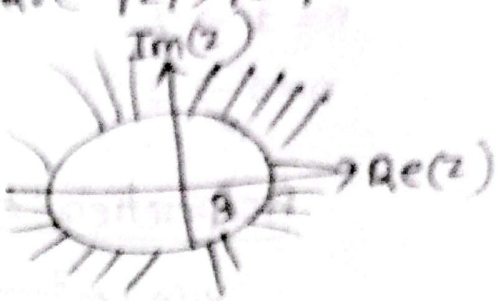
Same as $X_3(z) = \frac{4}{1-3z^{-1}}$ ROC $|z| > |3|$

$X(z) = \frac{3}{1-2z^{-1}} - \frac{4}{1-3z^{-1}}$: final ROC $|z| > 3$

ROC: 1 $|z| > 2$: 3/4,5,6

ROC: 2 $|z| > 3$: 4,5,6

→ final ROC $|z| > |3|$



Time shifting

$x(n) \xleftrightarrow{z} X(z)$

$x(n-k) \xleftrightarrow{z} z^{-k} X(z)$

Example অংশের #

$$\sum_{n=-\infty}^{\infty} x(n-k) z^{-n}$$

let $m = n-k$

$n = m+k$

$n = -\infty \quad m = -\infty - k = -\infty$

$n = \infty \quad m = \infty - k = \infty$

$$= \sum_{n=-\infty}^{\infty} x(m) z^{-(m+k)}$$

$$= \sum_{m=-\infty}^{\infty} x(m) z^{-m} \cdot z^{-k}$$

$$= z^{-k} X(z)$$ Ans

Problem: Using the time shifting property, find the z-transform of $x_2(n)$ $x_3(n)$ using $x_1(n)$

$x_1(n) = [1, 2, 5, 7, 0, 1]$ (↑ at 0, ↑ at 5)

$$X_1(z) = \sum_{n=0}^{5} x_1(n) z^{-n} = x_1(0) z^{0} + x_1(1) z^{-1} + x_1(2) z^{-2} + x_1(3) z^{-3} + x_1(4) z^{-4} + x_1(5) z^{-5}$$

$$= 1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + z^{-5}$$

ROC: Entire z plane except $z = 0$

DSP Part 1 Lec-3-6

$x_2[n] = \{1, 2, 5, 7, 0, 1\}$ (↑ at 1)

$x_2[n] \Rightarrow x_1(n+2)$

$x_2(z) \longrightarrow z^{2} X(z)$

$X_2(z) = z^{2}\left(1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + z^{-5}\right)$

$= z^{2} + 2z^{1} + 5 + 7z^{-1} + z^{-3}$

Ans

ROC: Entire z-plane except $z = 0$ and $z = \infty$

$x_3(n) = [0, 0, 1, 2, 5, 7, 0, 1]$

$x_3(n) \longrightarrow x_1(n-2)$

$x_3(z) \longrightarrow z^{-2} X(z)$

$X_3(z) = z^{-2} X(z)$

$= z^{-2}\left(1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + z^{-5}\right)$

$= z^{-2} + 2z^{-3} + 5z^{-4} + 7z^{-5} + z^{-7}$

ROC: Entire z plane except $z = 0$